

## Task:

You are working as a junior process engineer in a semiconductor company. The company plans to fabricate a low-power CMOS inverter (nMOS + pMOS on the same wafer) for a digital IC operating at 1.8 V. Your task is to design a feasible monolithic fabrication process flow, starting from a silicon wafer and ending at a packaged chip under Practical Constraints.

## 1. Objective and Problem Description

The objective of this work is to outline the fabrication process of a low-power CMOS inverter intended for digital integrated circuit applications operating at a supply voltage of 1.8 V. The inverter is composed of a complementary pair of n-channel MOSFET (nMOS) and p-channel MOSFET (pMOS) devices fabricated on a single silicon substrate.

## 2. CMOS Fabrication Process Flow

### Step 1: Silicon Wafer Selection and Cleaning

The process begins with a lightly doped n-type silicon wafer serving as the base substrate. To ensure a defect-free surface, the wafer undergoes a standard RCA cleaning procedure. This includes SC-1 cleaning to remove organic residues and particulate contamination, followed by SC-2 cleaning to eliminate metallic impurities.

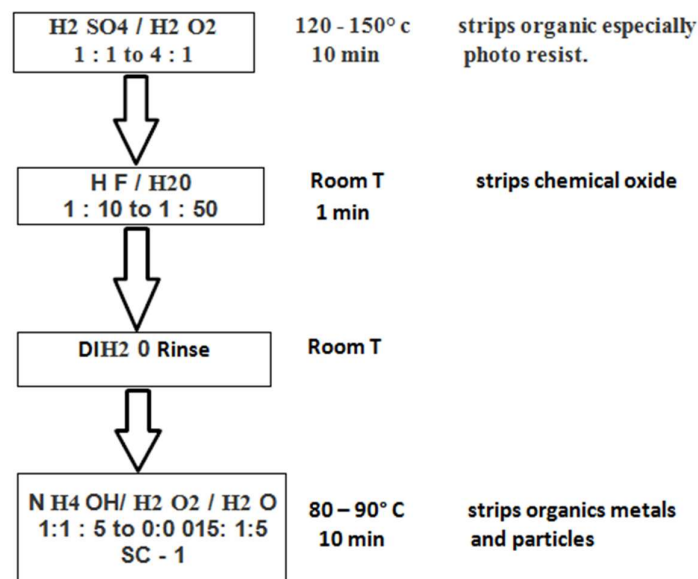


Fig1: RCA cleaning process

## Step 2: Device Isolation Using Field Oxide

To electrically separate neighboring devices, field isolation is implemented using a LOCOS (Local Oxidation of Silicon) technique. A thick silicon dioxide layer, typically around 400–500 nm, is thermally grown on non-active regions to prevent leakage currents and parasitic device interaction.

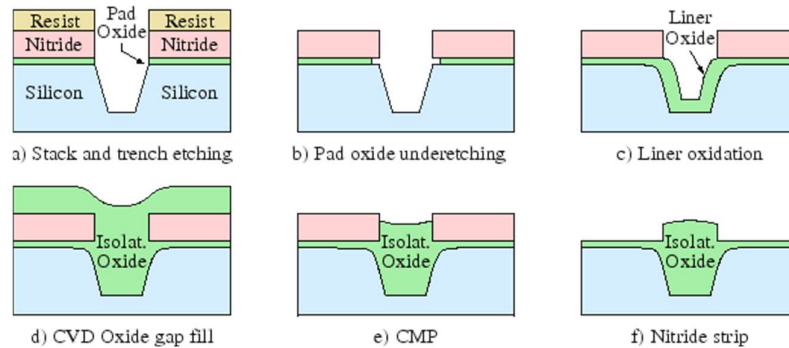


Fig2: Isolation technique step by step

## Step 3: Well Formation for nMOS Device

Photolithography is used to define the well regions. Donor dopants are introduced through ion implantation, followed by a high-temperature drive-in diffusion process to form a uniform p-well required for nMOS transistor fabrication.

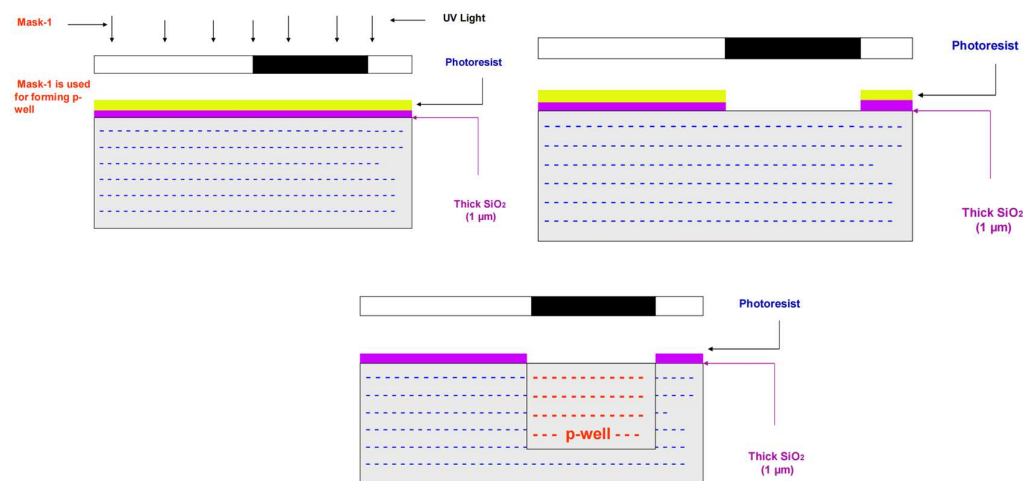


Fig3: Steps of formation of p-well

## Step 4: Gate Oxide Formation

After removing the thick oxide from the active areas, a thin gate oxide layer of approximately 8–10 nm is grown using thermal oxidation. This oxide thickness is suitable for 1.8 V operation and ensures proper gate control while minimizing leakage current.

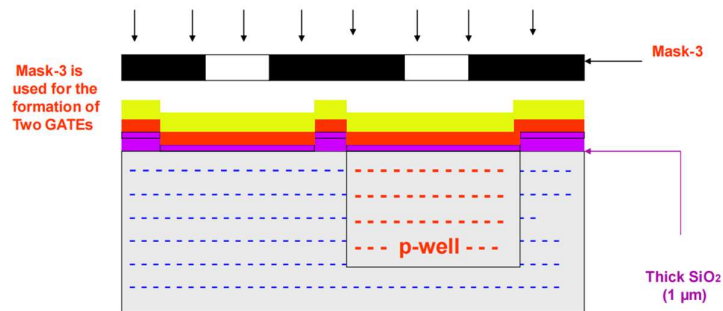
### Step 5: Polysilicon Gate Deposition

A polysilicon layer is deposited over the wafer using chemical vapor deposition (CVD). The polysilicon is subsequently doped to reduce its resistivity, making it suitable for use as the gate electrode for both nMOS and pMOS transistors.



### Step 6: Gate Pattern Definition

Photolithography steps including photoresist application, mask alignment, ultraviolet exposure, and development are carried out to define the gate geometry. The unwanted polysilicon is removed using anisotropic dry etching to achieve precise gate dimensions and sharp profiles.

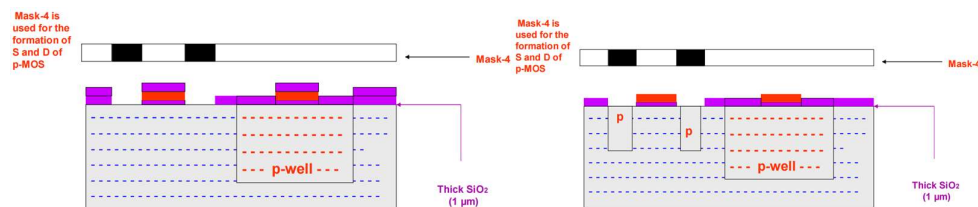


### Step 7: Source and Drain Formation

Separate masking steps are performed for source and drain implantation:

- Donor ions are implanted to form n<sup>+</sup> source and drain regions for the nMOS transistor.
- Acceptor ions are implanted to create p<sup>+</sup> source and drain regions for the pMOS transistor.

A brief thermal annealing process activates the dopants and repairs crystal lattice damage caused by ion implantation.



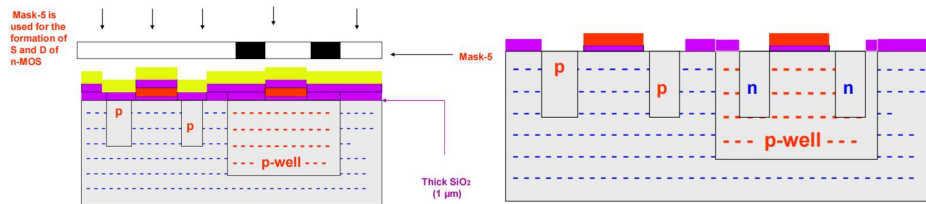
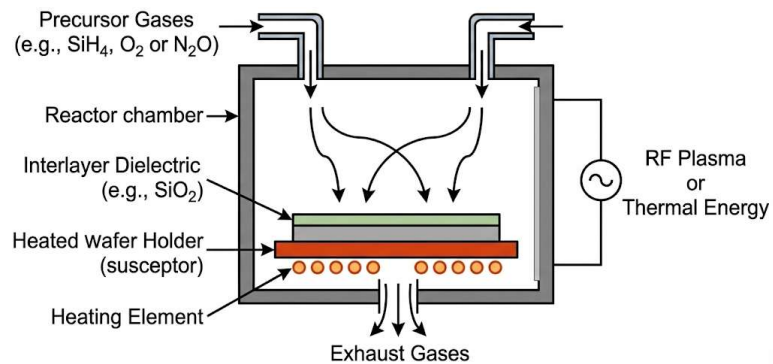


Fig4: Source and Drain formation.

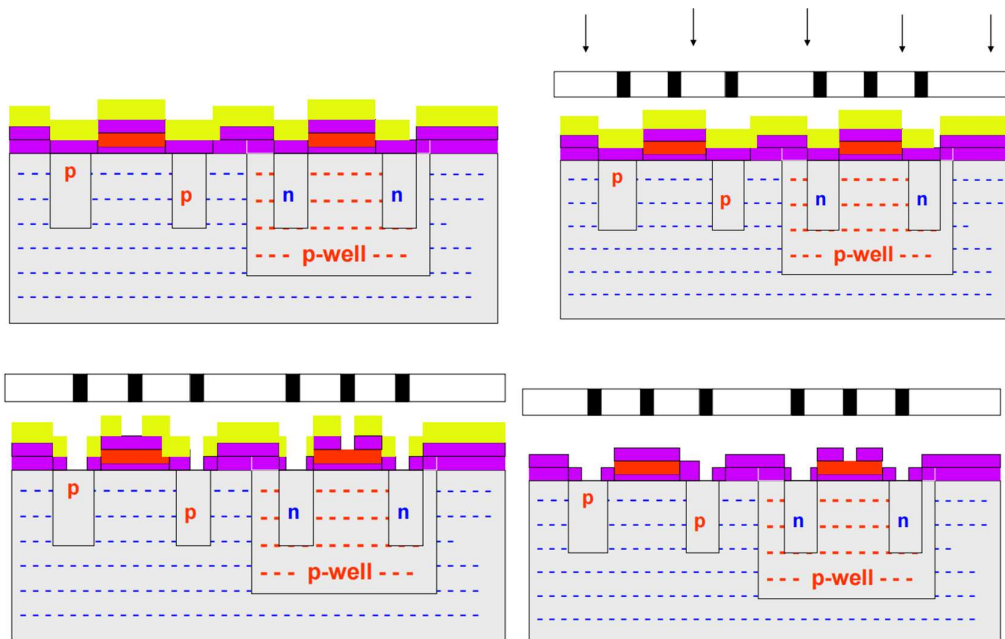
### Step 8: Interlayer Dielectric Deposition

An interlayer dielectric, typically silicon dioxide, is deposited using CVD. This insulating layer electrically separates the transistor structures from the upcoming metal interconnect layers.



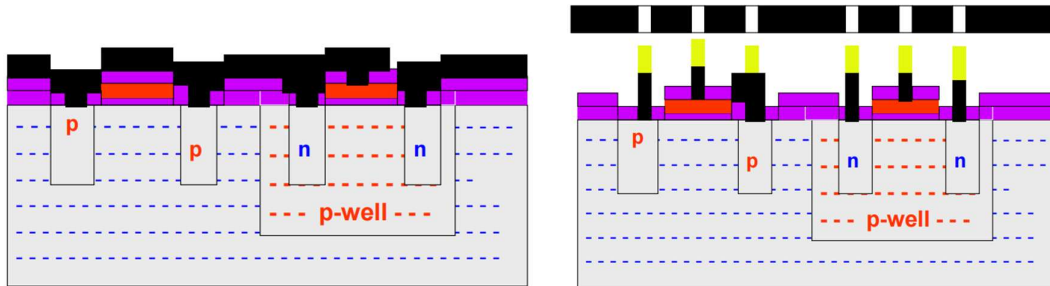
### Step 9: Contact Hole Formation

Using photolithography and selective wet etching, contact openings are created through the dielectric layer to expose the source, drain, and gate regions. Wet etching is selected for its excellent selectivity between silicon dioxide and silicon.



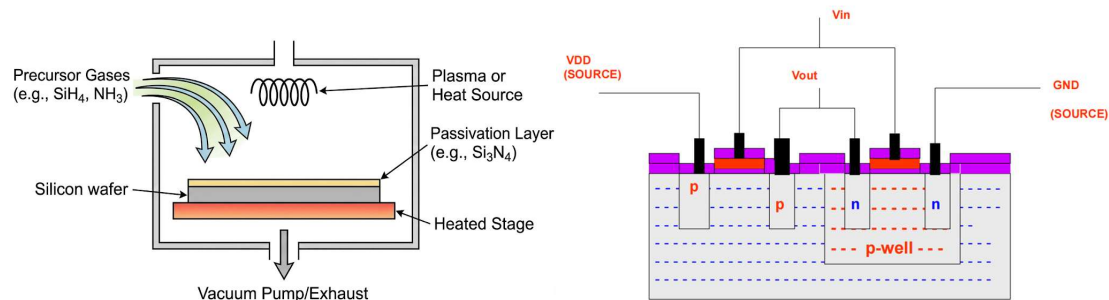
### Step 10: Metallization and Interconnection

An aluminum layer is deposited by sputtering to form electrical interconnections. The metal layer is patterned using photolithography and etching to connect the nMOS and pMOS transistors, thereby completing the CMOS inverter circuit.



### Step 11: Passivation Layer Deposition

A final passivation layer made of silicon dioxide or silicon nitride is deposited over the wafer. This layer protects the circuit from environmental factors such as moisture, dust, and mechanical stress.



### Step 12: Testing and Packaging

Electrical testing is carried out to verify the correct operation of the CMOS inverter. After testing, the wafer is diced into individual chips, wire bonding is performed, and the devices are packaged for practical use.

## 3. Design Justification

CMOS technology is selected due to its superior power efficiency and suitability for modern low-voltage digital circuits. Ion implantation is preferred over diffusion because it provides accurate control over dopant concentration and junction depth. LOCOS isolation offers an effective and straightforward method for device separation. The thin gate oxide ensures reliable transistor switching at low voltage, while aluminum metallization provides good conductivity and compatibility with standard fabrication processes.

#### **4. Conclusion**

This assignment presents a complete and realistic fabrication sequence for a low-power CMOS inverter using conventional semiconductor processing techniques. The proposed process flow meets electrical performance requirements, supports low-voltage operation, and maintains manufacturability. Therefore, it is well suited for basic digital IC fabrication and academic learning purposes.

